TEXORA · LECTURE SHEET

Fatigue, Flex Life and Fiber Ropes

Most ropes do not die in a single overload. They die by repetition: thousands of load cycles, bends over sheaves and rubbing contacts quietly consume their strength. This sheet covers fatigue and flex life, the fibre-rope materials, and the safety factors that keep working loads honest.

YM-4203 · TECHNICAL FIBRE & YARN · TOPIC 5 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. Fatigue — failure by repetition

Fatigue is the loss of strength, and eventual failure, caused by repeated loading at levels well below the single-pull breaking load. A yarn or rope that breaks at 100 kN in one pull may fail after thousands of cycles at 40 kN, or after far fewer at 60 kN. Three loading patterns matter for fibres and ropes:

- **Cyclic tensile fatigue** — the load rises and falls repeatedly between two tensile levels, as in a mooring line worked by waves or a lifting rope in repeated service.
- **Flex fatigue** — repeated bending, typically over a sheave or around a pin, which strains the outer fibres of the bend in tension and the inner fibres in compression on every pass.
- **Abrasion interplay** — fatigue rarely acts alone. Cyclic movement makes fibres and strands rub against each other (internal abrasion) and against hardware (external abrasion), and the fibrillated, damaged surface then fatigues faster. Treating abrasion and fatigue as separate problems understates real rope wear.

Two features distinguish fatigue from simple overload. First, damage is **cumulative and largely invisible**: a rope can retain its diameter and most of its appearance while having lost a serious fraction of its strength. Second, life is **statistical**: nominally identical specimens tested identically fail over a range of cycle counts, so fatigue results are reported as distributions and compared at a stated survival level, not as single numbers.

2. The S–N curve

Fatigue behaviour is summarised on an **S–N curve**: the cyclic stress (or load, as a percentage of breaking load) S plotted against the number of cycles to failure N, with N on a logarithmic axis because lives range from tens to millions of cycles.

- At high S, failure comes in few cycles — the curve starts high on the left.
- As S falls, life rises steeply: a modest reduction in working load buys a large multiple of life. This is the single most useful practical message of the S–N curve.
- Some materials show a flattening — an endurance limit below which life becomes very long. For fibre ropes in real service, abrasion, creep and environmental damage continue even at low cyclic loads, so a true “infinite life” plateau must never be assumed for a rope in use.

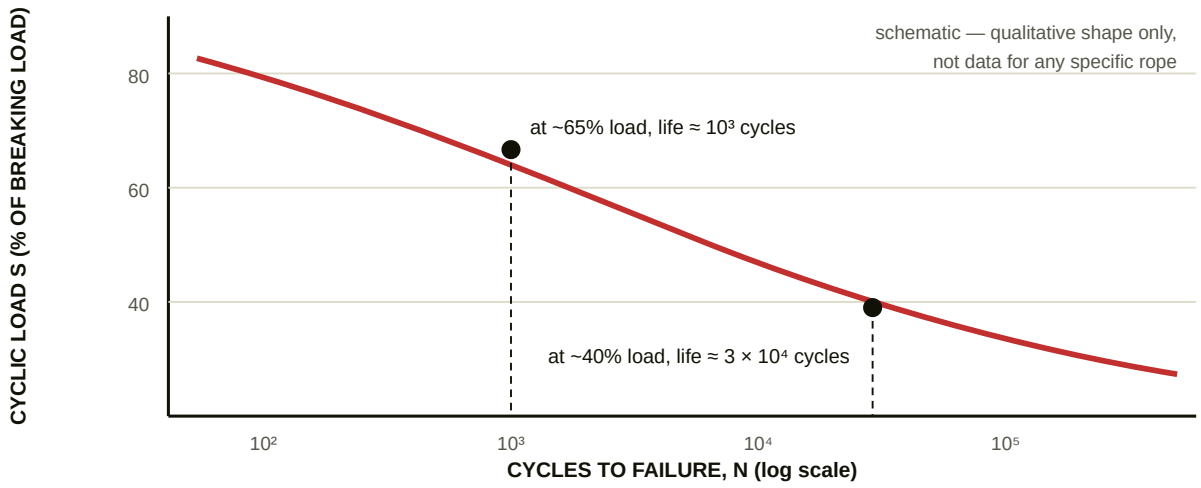


Figure 1 — Schematic S–N curve for a fibre rope. Lowering the cyclic load level multiplies the cycles to failure; the values shown illustrate the reading of the curve only.

2.1 Reading the curve — a qualitative example

Using Figure 1 as illustration only: at about 65% of breaking load a rope fails at roughly 10^3 cycles; at about 40% it survives roughly 3×10^4 cycles — about thirty times the life for a twenty-five-point load reduction. Real design uses tested S–N data for the specific rope in its actual environment.

3. Flex life and its testing

Flex life is the number of bending cycles a yarn, cord or rope survives under stated conditions. It depends on:

- **Bend severity** — the ratio of sheave (or pin) diameter D to rope diameter d. Small D/d means a sharp bend and short life; running rigging standards commonly demand generous D/d, and halving the sheave diameter can cost far more than half the flex life.
- **Tension during bending** — a bend under high load presses the strands together, raising internal friction, abrasion and heat.
- **Fibre bending behaviour** — fibres that tolerate repeated compression and kinking on the inside of the bend (nylon, polyester, UHMWPE) outlast brittle fibres; aramids, for all their tensile excellence,

are notably weak in repeated flexing and compression, which is why aramid ropes are used cautiously in bend-over-sheave service.

3.1 How flex life is tested

In a typical **flex-fatigue test** for yarns and cords, the specimen is bent repeatedly back and forth over a pin or blade of stated diameter, under a stated tension or bending angle, and the cycles to rupture are counted. Rope-level testing runs the rope cyclically over a sheave of stated D/d at a stated percentage of breaking load, recording cycles to failure and, in better programmes, the residual strength of survivors. Test reports must state fibre, construction, D/d, load level, bending angle or stroke, and environment — a flex-life number without those conditions cannot be transferred to another application.

> **KEY IDEA — HYSTERESIS HEATING**
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> Every load or bend cycle converts a small part of the work into heat inside the rope, because the loading and unloading curves do not coincide (hysteresis). At high cycle rates or high loads this internal heating can raise the core temperature enough to weaken or even melt synthetic fibres from the inside — a failure that shows a fused core in a rope whose surface looked serviceable. Cycle rate is therefore part of any fatigue specification.

4. Fibre ropes — materials compared

All property statements below are standard textbook descriptions of typical grades.

FIBRE	STRENGTHS FOR ROPE WORK	LIMITATIONS	TYPICAL ROPE USES
Nylon (polyamide)	Highest energy absorption: stretches (roughly 15–25% at break) and absorbs shock loads; good abrasion and fatigue behaviour	Loses roughly 10–15% strength when wet; sinks; UV degrades it over time	Mooring and towing where shock absorption matters, climbing (dynamic) ropes
Polyester	Strong, low stretch, excellent UV and wet performance, strength unchanged wet	Less shock absorption than nylon; sinks	General marine ropes, halyards, industrial slings
HMPE (UHMWPE)	Highest specific strength; floats; excellent abrasion and flex behaviour	**Creep** under sustained load; melts ~150 °C; slippery, needs correct splicing and hardware	High-performance mooring, towing, winch lines

Aramid	Very high strength and modulus, low stretch, heat resistant	Poor flex/compression fatigue; UV sensitive — needs a jacket; higher cost	Static lines, heat-exposed and precision-length applications
Polypropylene	Cheap, floats, chemically resistant, easy to splice	Lowest strength of the group, poor UV life unless stabilised, melts ~165 °C, creeps	Light utility, aquaculture, barrier ropes

The selection logic mirrors Topic 1: state the duty first. A rope that must absorb the snap of a towing load wants nylon's extension; the same extension is a fault in a halyard that must hold a sail's shape, where polyester or HMPE wins; a permanently loaded HMPE mooring must be assessed for creep, not just for breaking load.

5. How ropes degrade in service

- **Ultraviolet light** — breaks polymer chains at the surface; polypropylene and aramid are particularly sensitive, polyester the most resistant of the common rope fibres. Jackets and covers protect load-bearing cores.
- **Abrasion** — external (against decks, fairleads and seabeds) cuts fibres progressively inward; internal abrasion between strands powders fibre from inside during cyclic working. Chafe protection at contact points is a life-extender, not an accessory.
- **Creep** — sustained load produces continuing extension, worst in HMPE and polypropylene; it changes line lengths and tensions in permanent installations.
- **Hysteresis heating** — rapid, heavy cycling heats the core (Section 3.1).
- **Chemical and biological attack** — generally modest for synthetics, but acids damage nylon and alkalis damage polyester; natural-fibre ropes additionally rot.
- **Knots and bad practice** — a knot typically costs 30–50% of a rope's strength, and shock loading, undersized sheaves and twisted (hockled) laid rope all consume life that no inspection can restore.

Because fatigue damage is internal, rope management is a system: record history, inspect, retire on stated criteria, and never return a suspect rope to critical duty because it “still looks all right”.

6. Safety factors and working load limits

The **minimum breaking load (MBL)** is the lowest load at which a new rope of that specification is expected to break in test. Nobody works a rope at its MBL. The **working load limit (WLL)** is set by dividing by a **safety factor** chosen for the application — larger where people are at risk, loads are dynamic, or the environment degrades the rope:

Working load limit = minimum breaking load / safety factor

WORKED EXAMPLE — WORKING LOAD LIMIT

A rope has a minimum breaking load of 120 kN. The application requires a safety factor of 6.

WLL = 120 / 6 = **20 kN**.

The rope may therefore be assigned working loads up to 20 kN. Note what the factor is buying: allowance for fatigue and ageing, for dynamic peaks above the nominal load, for knots, abrasion and imperfect load sharing — the factor is consumed by real effects, not held as spare strength for extra load.

Safety factors are specified by the governing standard or regulation for each duty (lifting of goods, lifting of persons, mooring, climbing and so on) and those specified values — not personal judgement — must be used in real work. Personnel duties always demand substantially higher factors than goods-only duties.

Glossary

Fatigue — strength loss and eventual failure under repeated loading below the single-pull breaking load. **S–N curve** — cyclic load level plotted against cycles to failure on a logarithmic axis. **Flex life** — number of bending cycles survived under stated conditions. **D/d ratio** — sheave diameter divided by rope diameter; a measure of bend severity. **Hysteresis** — non-coincidence of loading and unloading curves, whose area is energy lost as heat. **Creep** — continuing extension under sustained load. **MBL** — minimum breaking load of a new rope in test. **WLL** — working load limit: MBL divided by the safety factor.

Key points

• Fatigue damage is cumulative, statistical and largely invisible: history and retirement criteria, not appearance, govern rope safety.

• The S–N curve's practical lesson: a modest cut in cyclic load buys a large multiple of life.

• Flex life collapses with small D/d, high tension and rapid cycling; aramid is strong in tension but weak in repeated flexing.

• Nylon absorbs shock, polyester resists UV and water, HMPE is strongest per weight but creeps, polypropylene is cheap and floats but weak and UV-sensitive.

• Degradation routes: UV, internal and external abrasion, creep, hysteresis heating, knots.

• WLL = MBL / safety factor, with the factor taken from the governing standard for the duty.

Practice

1. A lifting rope has an MBL of 180 kN. Calculate its WLL with a safety factor of 9, and again with a factor of 5. Explain why the factor for lifting persons must be the larger one and what real effects it covers.
2. Using the S–N idea qualitatively, explain why reducing a mooring line's peak cyclic load from about 65% to about 40% of breaking load changes its expected life from the order of thousands of cycles to the order of tens of thousands.
3. An aramid rope is proposed for a winch that bends it over a small sheave thousands of times a week. Assess the proposal and recommend a better fibre, with reasons.
4. List four inspection findings that should trigger retirement of a synthetic rope even though it has never been overloaded, and connect each finding to its degradation mechanism.

Answer key

1. WLL at factor 9 = 180 / 9 = **20 kN**. WLL at factor 5 = 180 / 5 = **36 kN**. Personnel duty uses the larger factor because failure consequences are fatal rather than economic, and because the factor must absorb fatigue and ageing loss, dynamic peaks, knot and termination losses, abrasion and imperfect load sharing — all of which act whether or not anyone is riding the load; only the consequence changes.
2. Fatigue life falls steeply as cyclic load rises — the S–N curve is not a straight proportionality. Near 65% of breaking load each cycle does heavy damage and failure comes at roughly 10^3 cycles on the schematic curve; at about 40% each cycle does far less damage, so the rope accumulates roughly 3×10^4 cycles before failure. The gain comes from the shape of the curve: load reduction buys life at a multiplying, not an adding, rate.
3. Reject the proposal: aramid's weakness is exactly this duty — repeated bending puts its fibres into compression on the inside of the bend, where aramid fibrillates and loses strength rapidly, and a small sheave makes the bend severe. Recommend **HMPE** for its excellent flex and abrasion behaviour and specific strength (checking creep and temperature), or **polyester** where lower cost and proven sheave service matter more than weight — in both cases with a generous D/d.
4. (1) **Local diameter reduction / necking** — internal fibre breakage from fatigue and internal abrasion. (2) **Glazed, fused surface or core** — hysteresis heating or friction melting. (3) **Powdery fibre dust inside the lay** — internal abrasion between strands. (4) **Stiff, boardy handling with broken surface fibres over long lengths** — advanced UV and flex-fatigue degradation. Any one of these means internal strength has gone even if the rope once passed a visual glance.
